Fool’s Descent - Database Modeling Exercise for the Challenge

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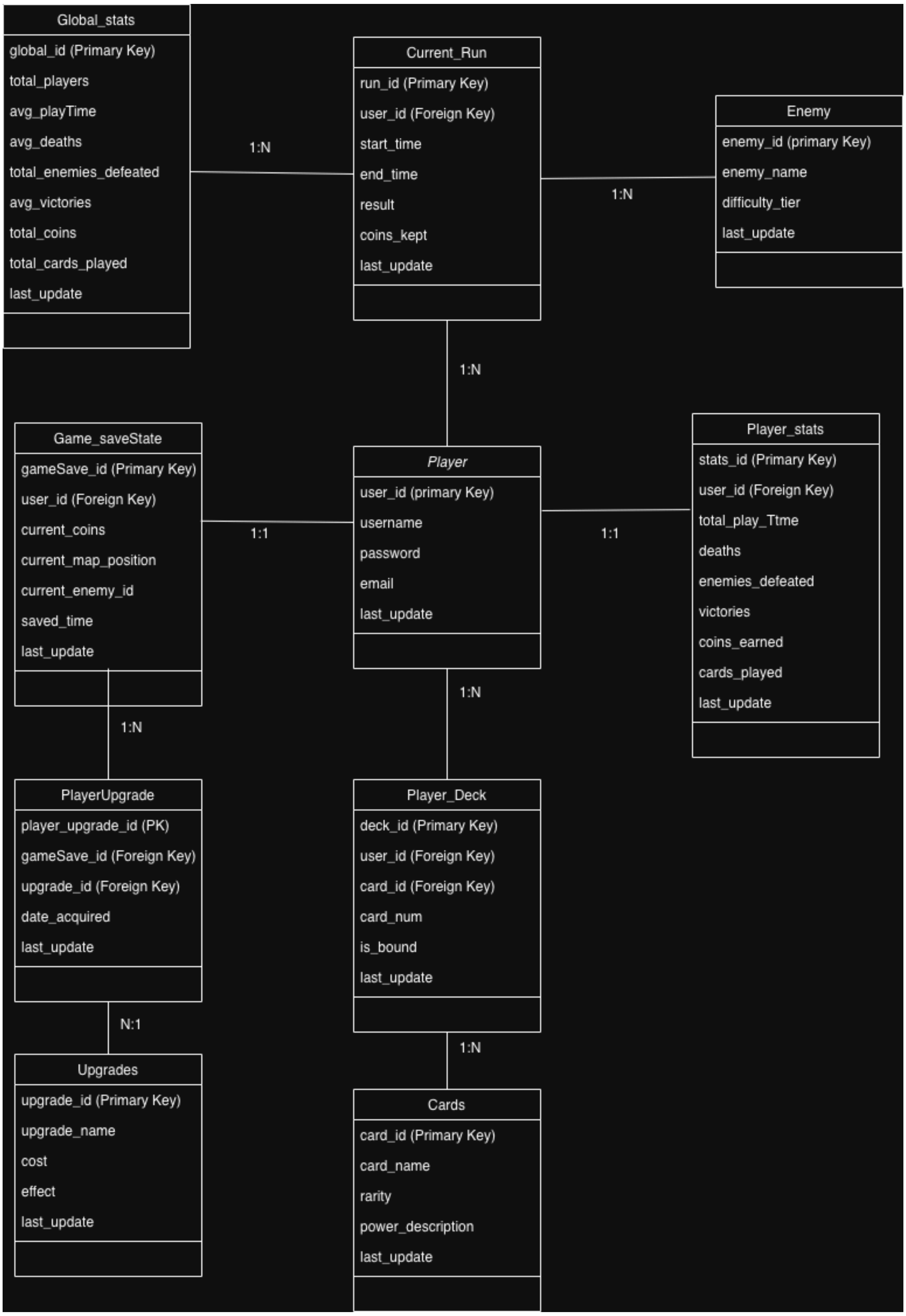

Justification for each table:

Player

This table stores the core identity of each player, including their username, password and email. It is essential because the game includes a login system and allows users to save progress, continue their descent later and maintain persistent statistics across multiple runs.

Player_stats

This table stores the personal performance statistics of each player, such as total play time, deaths, victories, enemies defeated, coins earned and cards played. It is important because it allows players to track their long-term progression and supports the statistics screen described in the game design.

Global_stats

This table stores aggregated data across all players, such as total players, average play time, total enemies defeated and total cards played. Its purpose is to provide global statistics that help evaluate game balance, player behavior and overall community progression.

Current_Run

This table stores the information of each run or descent attempt made by the player, including start time, end time, result, and coins kept after the run ends. It is important because the game is a roguelike, meaning each run is unique and must be recorded separately to preserve progression and historical performance.

Game_saveState

This table stores the current saved progress of the player during an active run, including coins, current map position, current enemy and save time. It is necessary because the game includes a “Continue Descent” button, allowing players to leave and later resume exactly where they stopped.

Enemy

This table stores all enemy definitions in the game, including their name and difficulty tier (Easy, Mid Difficulty, Hard Difficulty, Final Boss). It is important because enemies are central to progression, rewards, and difficulty scaling and each run depends on randomized enemy encounters.

Cards

This table stores all playable cards in the game, including their name, rarity, and effect description. It is essential because the entire gameplay loop revolves around card management, where players use cards to manipulate fate and survive duels.

Player_Deck

This table stores the cards the player currently owns, including quantity and whether a card is permanently bound. It is important because each player has a personalized deck that changes over time depending on victories, upgrades and strategic decisions.

Upgrades

This table stores all upgrade options available during the map progression, such as Card Binding and Life Extension, including their cost and effect. It is necessary because upgrades create long-term strategic progression and allow players to improve future runs.

PlayerUpgrade

This table acts as an intermediate relationship between Game_saveState and Upgrades, recording which upgrades a player has acquired and when. It is important because multiple players can obtain the same upgrade and each saved game can contain multiple upgrades, making this many-to-many relationship necessary for proper tracking.